DESIGN OF MACHINE ELEMENTS

BMEE301L

WINTER SEMESTER 2025-26

DESIGN AND DEVELOPMENT OF A SPRING-BASED WALL-PRESSED IN-PIPE INSPECTION ROBOT

PROJECT REPORT

TEAM MEMBERS:

-RISHIKA R 23BMH1050

-CHRISTS DARYN P 23BMH1126

Design and Development of a Spring-Based Wall-Pressed In-Pipe Inspection Robot

Abstract

This paper presents the design and development of a wall-pressed in-pipe inspection robot capable of operating in pipelines with diameters ranging from 550 mm to 600 mm. The robot utilizes a three-arm configuration arranged at 120°, with each arm incorporating a rocker-bogie mechanism for improved adaptability to uneven pipe surfaces. A spring-based radial expansion mechanism is employed to generate normal force for traction. Detailed force, torque, and spring design calculations are carried out to ensure reliable locomotion. Additionally, finite element analysis of the spring assembly is performed using SOLIDWORKS Simulation to validate structural integrity. The proposed system provides a simple, cost-effective, and efficient solution for pipeline inspection.

Keywords— In-pipe robot, wall-pressed mechanism, spring design, rocker-bogie, traction, FEA.

I. INTRODUCTION

Municipal water transmission mains (500–800 mm diameter) are critical pipelines supplying cities but suffer from biofilm, scaling, sediment buildup, and corrosion over time.

Current inspection methods require shutdown, dewatering, or have limited capability, making them costly, unsafe, and inefficient.

The proposed in-pipe robot operates without full shutdown, using a self-centering mechanism to navigate 550–600 mm pipes.

It combines a rotating brush and water jet system to clean deposits while moving through the pipeline.

An onboard camera enables real-time inspection of internal conditions like cracks, corrosion, and blockages.

This solution reduces inspection costs significantly while improving safety, efficiency, and maintenance planning for Indian water infrastructure.

II. SYSTEM DESIGN AND CONFIGURATION

The robot consists of a cylindrical central body with three arms symmetrically arranged at 120 degrees. Each arm is equipped with three wheels connected through a rocker-bogie mechanism, resulting in a total of nine wheels. The wheels are driven by DC geared motors to provide locomotion.

The robot incorporates a spring-based radial expansion mechanism that exerts outward force on the arms, ensuring continuous contact with the pipe walls. This design enables the robot to maintain traction even in varying pipe diameters and irregular surfaces.

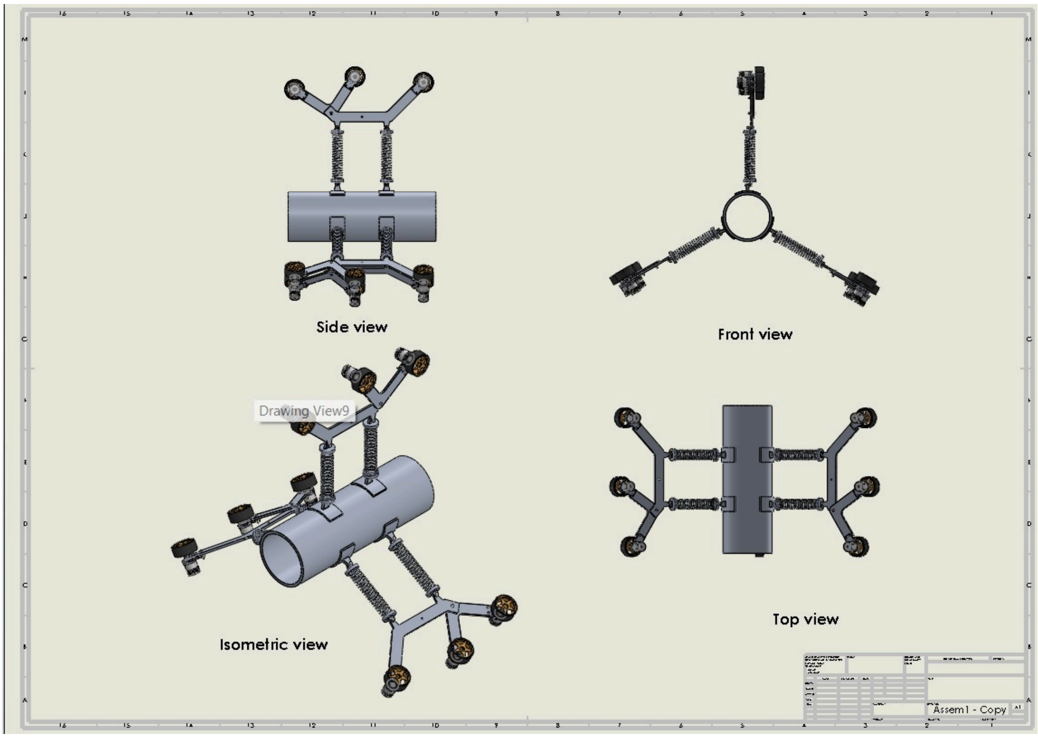


Fig. 1. Orthographic views of the in-pipe inspection robot (side, front, top, and isometric views).

III. WORKING PRINCIPLE

The robot operates based on friction-driven locomotion. The springs generate a normal force that presses the wheels against the pipe walls. The motors rotate the wheels, producing traction force given by:

$$F_{traction}=\mu F_{normal}$$

For motion to occur, the traction force must overcome resistive forces such as rolling resistance and gravitational forces in inclined pipes.

IV. FORCE AND TORQUE ANALYSIS

The total weight of the robot is calculated as:

$$W = mg = 4.5 \times 9.81 = 44NW$$

The rolling resistance is:

$$F_{rolling} = 0.05 \times 44 = 2.2N$$

The slope force for a 10° inclination is:

$$F_{slope} = 44\sin 10° \approx 7.6N$$

Thus, total force:

$$F_{required} = 10N$$

Applying safety factor:

$$F_{design} = 30N$$

Torque required:

$$T = 30 \times 0.035 = 1.05Nm$$

Torque per motor:

$$T_{motor} \approx 1Nm$$

Bending Stress Calculation for Arm

To evaluate the structural integrity of the robot arm, a bending stress analysis is performed. The total load acting on the system is calculated based on the mass of 1.5 kg, resulting in a weight of 14.7 N. Since the load is distributed across three wheels, the force acting on each wheel is approximately 4.9 N. Considering the arm length of 90 mm, the bending moment is calculated as 441 N·mm.

The cross-section of the arm is assumed to be rectangular with a width of 20 mm and thickness of 3.5 mm. The moment of inertia is calculated as 71.46 mm^4, and the distance from the neutral axis is 1.75 mm. Using the bending stress equation, the maximum stress induced in the arm is approximately 11 MPa.

This value is significantly lower than the material yield strength, indicating that the arm structure is safe under the given loading conditions.

Given:

$m = 1.5$ kg

$g = 9.81$ m/s^2

$W = m \times g = 1.5 \times 9.81 = 14.7$ N

Number of wheels = 3

Arm length $L = 90$ mm

Thickness $t = 3.5$ mm

Width $b = 20$ mm

Step 1: Load per wheel

$F = W / 3 = 14.7 / 3 = 4.9$ N

Step 2: Bending moment

$M = F \times L = 4.9 \times 90 = 441$ N·mm

Step 3: Moment of inertia

$I = (b \times t^3) / 12$

$I = (20 \times 3.5^3) / 12$

$I = (20 \times 42.875) / 12 = 71.46$ mm^4

Step 4: Distance from neutral axis

$c = t / 2 = 3.5 / 2 = 1.75$ mm

Step 5: Bending stress

$\sigma = (M \times c) / I$

$\sigma = (441 \times 1.75) / 71.46$

$\sigma \approx 10.8$ MPa

Final:

$\sigma \approx 11$ MPa

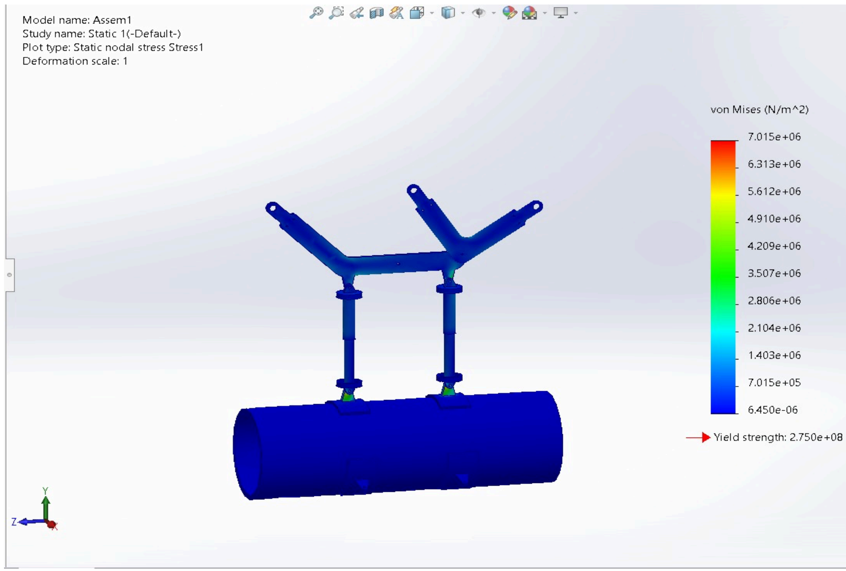


Fig. 2. Von Mises stress distribution of the robot arm assembly under static loading, showing maximum stress concentration at joint regions with peak value of 7.015 MPa.

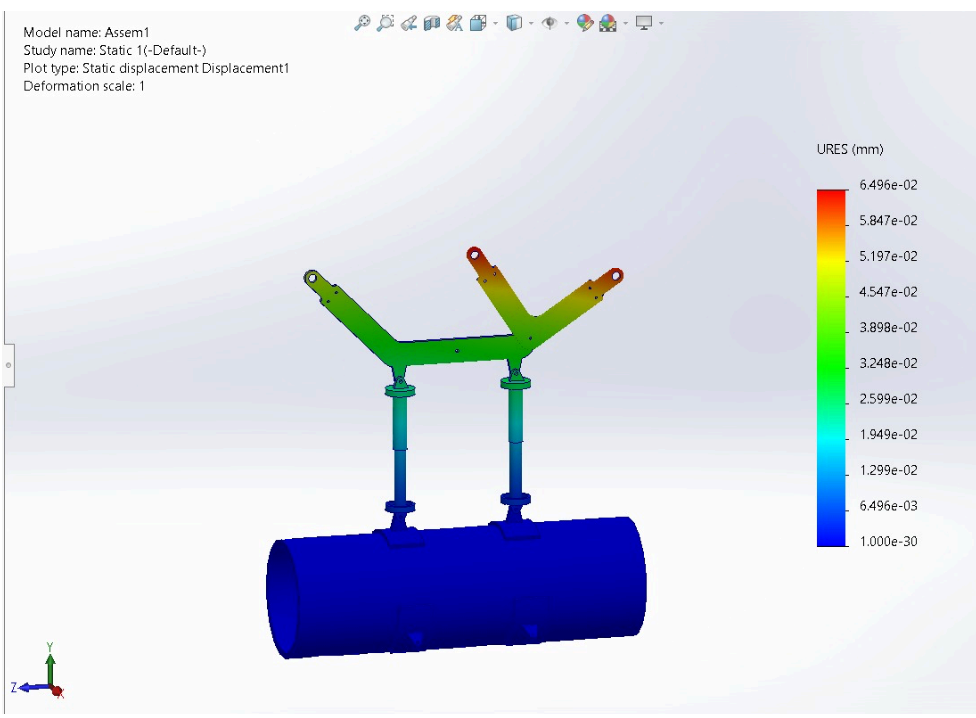


Fig. 3. Resultant displacement distribution of robot arm assembly under static loading.

V. SPRING DESIGN CALCULATION

The spring parameters are selected as follows:

$Lo=100\ mm$

$\delta=60mm$

$Do=20\ mm$

$d=1.5\ mm$

$D=Do-d$

$\quad=20-1.5$

$\quad=18.5mm$

The spring index is:

$$C = D/d = 12.3$$

Wahl correction factor:

$$K=((4C-1)/(4C-4))+(0.615/C)=1.1164$$

Shear stress equation:

$$\tau=(K8PC)/(\pi d^2)$$

$$450=(1.1164\times8\times P\times12.3)/(\pi\times1.52)$$

$$P=29N$$

For allowable stress τ=450 MPa:

$$P = 29N$$

Spring stiffness:

$$P/\delta = 29/60 = 0.4825 \text{ N/mm}$$

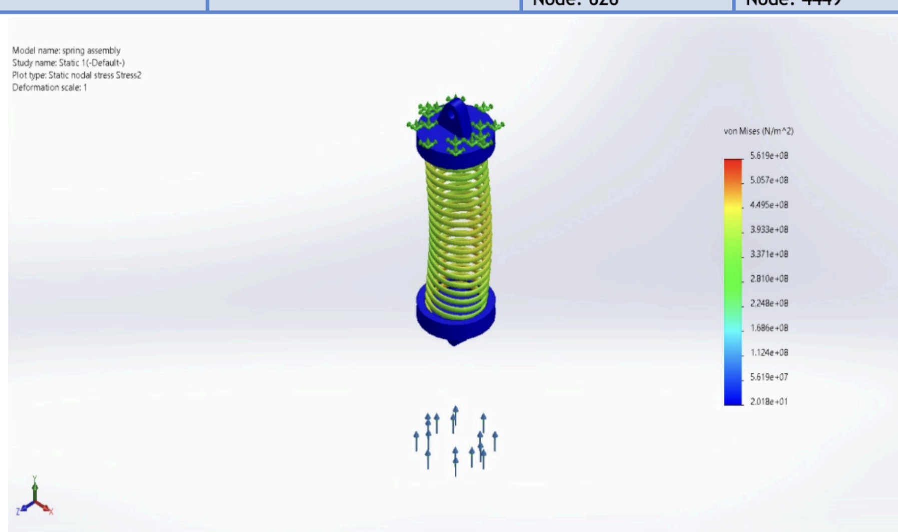
Theoretical stiffness:

$$k_{act} = Gd^4/8D^3N$$

$$= (81370 \times 1.5^4)/(8 \times 18.5^3 \times 20)$$

$$= 0.4066 \text{ N/mm}$$

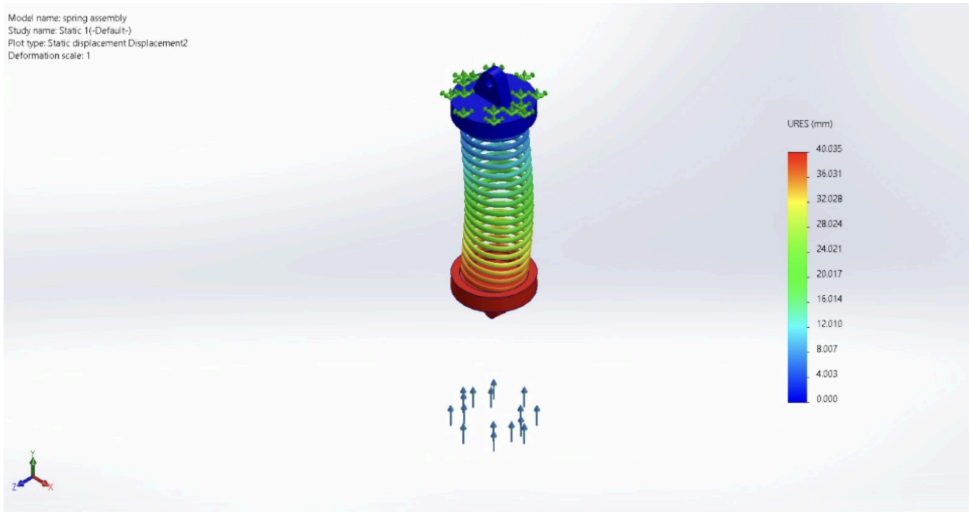
Name	Type	Min	Max
Stress2	VON: von Mises Stress	2.018e+01N/m^2 Node: 626	5.619e+08N/m^2 Node: 4449



spring assembly-Static 1-Stress-Stress2

Fig. 4. Von Mises stress distribution of spring assembly.

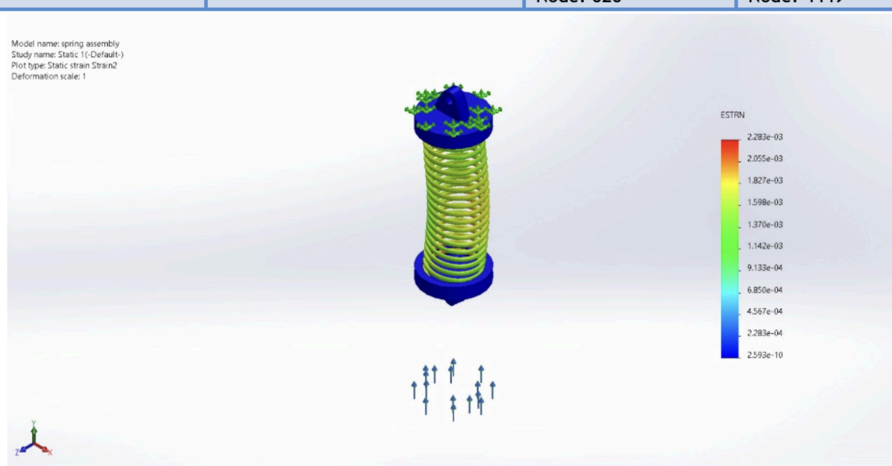
Name	Type	Min	Max
Displacement2	URES: Resultant Displacement	0.000mm Node: 1	40.035mm Node: 1172



spring assembly-Static 1-Displacement-Displacement2

Fig. 5. Displacement plot of spring under load.

Name	Type	Min	Max
Strain2	ESTRN: Equivalent Strain	2.593e-10 Node: 626	2.283e-03 Node: 4449



spring assembly-Static 1-Strain-Strain2

Fig. 6. Equivalent strain distribution of the spring assembly under static loading conditions.

from analysis

- σ_{max}=224 MPa
- σ_y=275 MPa

VI. FINITE ELEMENT ANALYSIS

The spring assembly was analyzed using SOLIDWORKS Simulation under static loading conditions. A load of 28 N was applied, and the system was constrained appropriately.

The results indicate a maximum von Mises stress of approximately 168 MPa, which is below the material yield strength of 275 MPa.

The factor of safety is:

$$FOS = 275/168 = 1.6$$

Maximum displacement observed is approximately 40 mm, which is within the allowable deflection range.

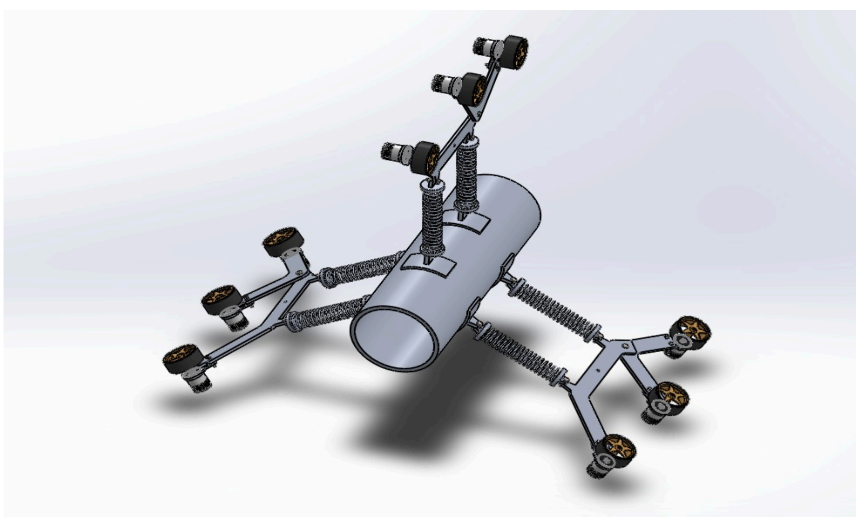
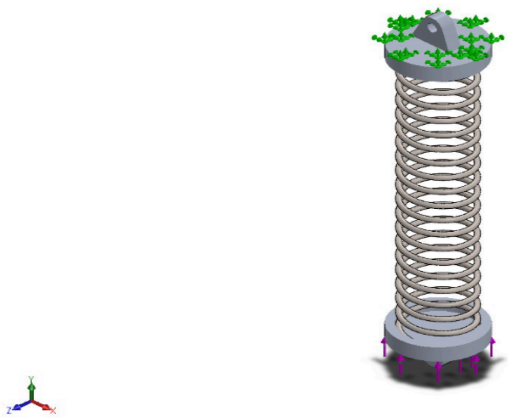


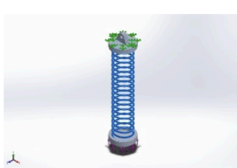
Fig. 7. 3D CAD model of the in-pipe inspection robot showing spring-based radial expansion mechanism.

VII. SIMULATION RESULTS

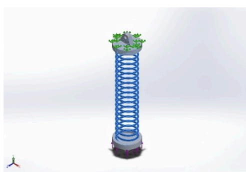
Model Information



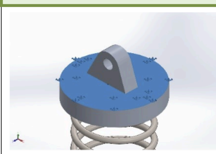
Model name: spring assembly
Current Configuration: Default

Solid Bodies			
Document Name and Reference	**Treated As**	**Volumetric Properties**	**Document Path/Date Modified**
Boss-Extrude3 	Solid Body	Mass:0.00486766 kg Volume:1.80284e-06 m^3 Density:2,700 kg/m^3 Weight:0.047703 N	C:\Users\Rishika\Desktop\6th sem\dome\project\spring attacher.SLDPRT Apr 15 01:02:35 2026
Boss-Extrude3 	Solid Body	Mass:0.00486766 kg Volume:1.80284e-06 m^3 Density:2,700 kg/m^3 Weight:0.047703 N	C:\Users\Rishika\Desktop\6th sem\dome\project\spring attacher.SLDPRT Apr 15 01:02:35 2026
Cut-Extrude2	Solid Body	Mass:0.0156805 kg Volume:2.03658e-06 m^3 Density:7,699.41 kg/m^3 Weight:0.153669 N	C:\Users\Rishika\Desktop\6th sem\dome\project\spring .SLDPRT Apr 14 21:41:19 2026
			

Material Properties

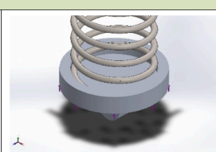
Model Reference	Properties	Components
	Name: 6061-T6 (SS) Model type: Linear Elastic Isotropic Default failure criterion: Unknown Yield strength: 2.75e+08 N/m^2 Tensile strength: 3.1e+08 N/m^2 Elastic modulus: 6.9e+10 N/m^2 Poisson's ratio: 0.33 Mass density: 2,700 kg/m^3 Shear modulus: 2.6e+10 N/m^2 Thermal expansion coefficient: 2.4e-05 /Kelvin	SolidBody 1(Boss-Extrude3)(spring attacher-1), SolidBody 1(Boss-Extrude3)(spring attacher-3)
Curve Data:N/A		
	Name: Alloy Steel Model type: Linear Elastic Isotropic Default failure criterion: Max von Mises Stress Yield strength: 6.20422e+08 N/m^2 Tensile strength: 7.23826e+08 N/m^2 Elastic modulus: 2.1e+11 N/m^2 Poisson's ratio: 0.28 Mass density: 7,700 kg/m^3 Shear modulus: 7.9e+10 N/m^2 Thermal expansion coefficient: 1.3e-05 /Kelvin	SolidBody 1(Cut-Extrude2)(spring-1)
Curve Data:N/A		

Loads and Fixtures

Fixture name	Fixture Image	Fixture Details
Fixed-2		Entities: 1 face(s) Type: Fixed Geometry

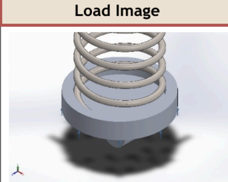
Resultant Forces

Components	X	Y	Z	Resultant
Reaction force(N)	1.15545e-05	-16.8567	2.99886e-06	16.8567
Reaction Moment(N.m)	0	0	0	0

Fixture name	Fixture Image	Fixture Details
On Flat Faces-1		Entities: 1 face(s) Type: On Flat Faces Translation: ---, ---, -40 Units: mm

Resultant Forces

Components	X	Y	Z	Resultant
Reaction force(N)	0	-0.0829034	0	0.0829034
Reaction Moment(N.m)	0	0	0	0

Load name	Load Image	Load Details
Force-1		Entities: 1 face(s) Type: Apply normal force Value: 28 N

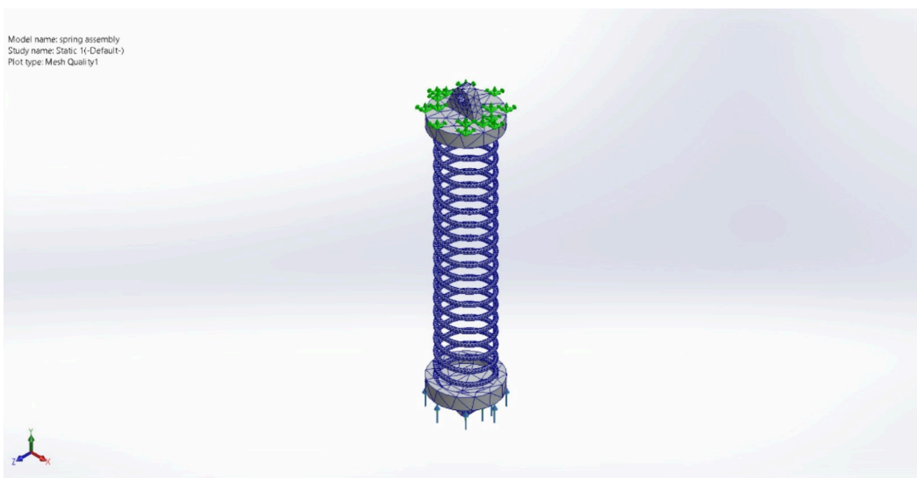
Mesh information

Mesh type	Solid Mesh
Mesher Used:	Blended curvature-based mesh
Jacobian points for High quality mesh	16 Points
Maximum element size	8.53367 mm
Minimum element size	1.00988 mm
Mesh Quality	High
Remesh failed parts independently	Off
Reuse mesh for identical parts in an assembly (Blended curvature-based mesher only)	Off

Mesh information - Details

Total Nodes	37782
Total Elements	16674
Maximum Aspect Ratio	6.8894
% of elements with Aspect Ratio < 3	96.8
Percentage of elements with Aspect Ratio > 10	0
Percentage of distorted elements	0
Time to complete mesh(hh;mm;ss):	00:00:44
Computer name:	

Mesh quality plots



Resultant Forces

Reaction forces

Selection set	Units	Sum X	Sum Y	Sum Z	Resultant
Entire Model	N	1.15545e-05	-16.9396	2.99886e-06	16.9396

Reaction Moments

Selection set	Units	Sum X	Sum Y	Sum Z	Resultant
Entire Model	N.m	0	0	0	0

VIII. RESULTS AND DISCUSSION

The robot design ensures stable locomotion within pipes of varying diameters. The spring-based radial expansion provides sufficient normal force for traction, while the rocker-bogie mechanism enhances adaptability to uneven surfaces. The finite element analysis confirms that the spring operates within safe stress limits. The design achieves a balance between simplicity, efficiency, and performance.

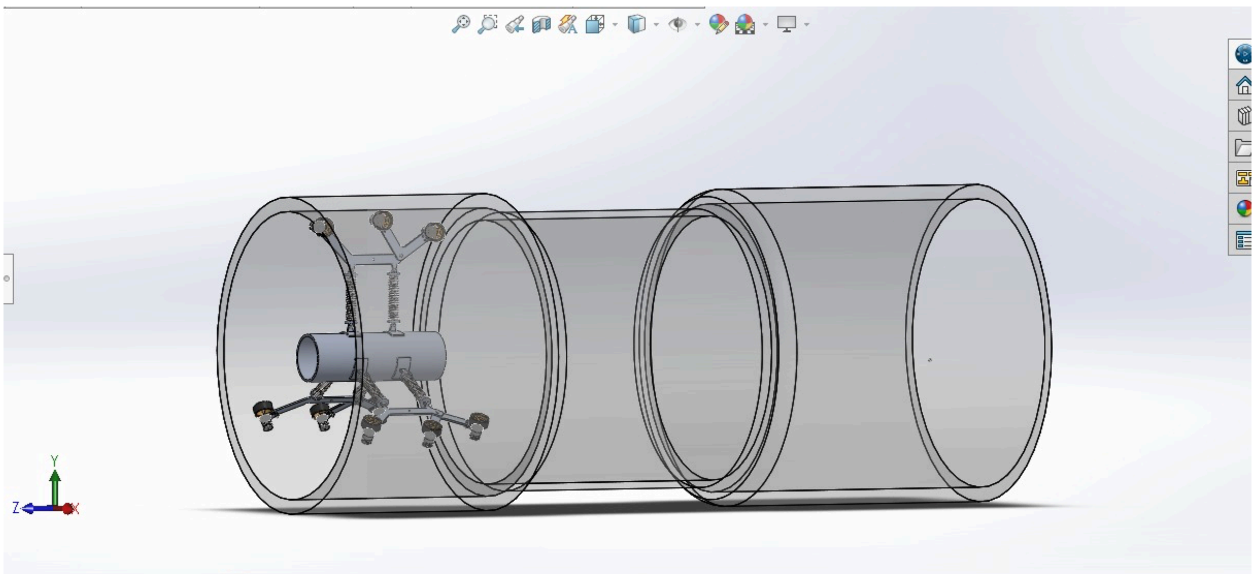


Fig. 8. isometric CAD assembly view of an in-pipe robot positioned inside a transparent, multi-section pipeline.

IX. APPLICATION AND JUSTIFICATION

Municipal water transmission mains are critical infrastructure responsible for supplying treated water across cities. In India, these pipelines typically range from 500 mm to 800 mm in diameter and extend over long distances. Over time, their internal condition deteriorates due to biofilm formation, mineral scaling, sediment accumulation, and corrosion, leading to reduced flow efficiency, contamination, and significant water losses.

Existing inspection methods such as manual entry, CCTV systems, and acoustic detection have major limitations, including the need for pipeline shutdown, safety risks, and inability to assess internal conditions effectively. These challenges make regular inspection difficult and costly.

The proposed in-pipe inspection robot addresses these issues by enabling inspection and cleaning without complete pipeline shutdown under low-pressure conditions. The three-arm spring-loaded mechanism ensures stable movement within pipes of 550–600 mm diameter, even with internal irregularities. The rotating brush system removes deposits, while water jets flush out debris. An onboard camera provides real-time monitoring of internal pipe conditions.

This solution is particularly relevant for Indian infrastructure, where ageing pipelines require efficient and cost-effective inspection. Compared to conventional methods costing ₹15–25 lakhs per kilometre, the proposed system reduces the cost to ₹3–5 lakhs per kilometre. For large pipeline networks, this results in substantial economic savings while improving safety and operational efficiency.

In summary, the proposed robot offers a practical, safe, and economical solution for inspection and maintenance of municipal water transmission mains, aligning with national initiatives such as the Jal Jeevan Mission.

X. CONCLUSION

A spring-based wall-pressed in-pipe inspection robot has been successfully designed and analyzed. The system demonstrates effective traction, adaptability, and structural safety within pipelines of varying diameters. The analytical (manual) bending stress calculated for the robot arm is approximately 11 MPa, while the finite element analysis (FEA) result shows a maximum stress of about 7 MPa. Both values are significantly lower than the material yield strength of 275 MPa, confirming that the structure operates well within safe limits.

The spring was theoretically designed for a deflection of 60 mm; however, simulation results indicate an achieved displacement in the range of 40–50 mm under practical loading conditions. This variation is attributed to realistic constraints such as boundary conditions and load distribution, but the obtained displacement remains sufficient to ensure proper radial expansion and contact with the pipe walls.

In terms of deformation, the maximum displacement obtained from simulation is approximately 6.496×10^{-2}mm, indicating minimal deformation and high structural stiffness. The close agreement between analytical calculations and simulation results validates the design methodology and assumptions used.

The use of passive spring-based radial expansion ensures continuous contact with the pipe walls while reducing mechanical complexity and energy consumption. This enhances the reliability and efficiency of the system in real-world operating conditions. Overall, the design satisfies both strength and stiffness requirements, making it suitable for practical implementation in in-pipe inspection applications. Future improvements may include the integration of active expansion mechanisms and advanced sensing technologies to further enhance performance and adaptability.